

Exercise 02

SQL Aggregate

Functions and Operations

- 1 List all distinct departments in the students table.

```
SELECT DISTINCT department  
FROM student-table;
```

Answer in a table format.

Department
IT
HR HR
Finance

- 2 Get the average of students per department

```
SELECT department,  
AVG (age) AS avg-age
```

```
FROM Students
```

```
GROUP BY department;
```

avg-age
21,6

Answer in table format

Department	avg-age
IT	20,5
HR	22
Finance	23

we group by a none aggregate columns. Hence the use of the department column.

3 Show departments with more than 1 student.

```
SELECT department, COUNT (*) AS total employees students
FROM employees emp students
GROUP BY department
HAVING COUNT (*) > 1;
```

Answer in a table format

department	total ^{students} employees
IT	2

4 Get all students whose age is between 21 and 23

```
SELECT *
```

```
FROM students
```

```
WHERE age BETWEEN 21 AND 23;
```

Answer in table format

Student_id	name	Age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5 List all students in the IT or HR department who are older than 21

```
SELECT students_id, department, name, age
FROM students
WHERE department IN ("IT", "HR")
AND age > 21;
```

Answer in a table format

Student_id	department	name	age
2	HR	Bob	22
5	HR	Eve	22

6 Show total credits per department, only for departments with more than 5 total credits.

```
SELECT department, SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5;
```

Answer in a table format

department	total_credits
IT	11

7 List all courses that do not have 4 credits

```
SELECT      department
            SUM (credits) AS total_credits
FROM        courses
GROUP BY    department
HAVING      SUM (credits) > 5;
```

Answer in table format

department	total_credits
IT	11

8 List all courses that do not have 4 credits

```
SELECT      course_id, course_name, department,
            credits
FROM        courses
WHERE       credits <> 4;
```

Answer in table format

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8 Show the top 3 courses by credits in descending order.


```

SELECT      course_id, course_name, credits
FROM        courses
ORDER BY    credits DESC
LIMIT 3;

```

Answer in table format

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

9 SELECT MAX(grade) AS max_grade,
 MIN(grade) AS min_grade,
 AVG(grade) AS avg_grade
 FROM enrollments

Answer in table format

max_grade	min_grade	avg_grade
90	78	84.6

10 Count how many enrollments exists per course.

```

SELECT      course_id,
COUNT(*) AS enrollment_count
FROM        enrollments
GROUP BY    course_id

```

Answer in table format

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

Find total salary and total bonus per department

```

SELECT      department
SUM (salary) AS total_salary,
SUM (BONUS) AS total_bonus,
FROM        salaries
GROUP BY    department ;
  
```

Answer in a table format

Name	Department	Salary	Bonus
Tom	IT	60000	5000
Jerry	HR	55000	4000
Spike	Finance	70000	6000
Tyke	IT	62000	5500
Buffy	HR	54000	3500

Final Output Table

Department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

12 Show departments where average salary is above 55,000,

SELECT department,
 AVG (salary) AS avg-salary

FROM salaries
GROUP BY department
HAVING AVG (salary) > 55000;

Answer in a table format

Department	avg-salary
IT	65000
Finance	70000